Synthetic-Data-Driven Public Adoption Prediction of Connected and Autonomous Vehicles: A Hybrid Explainable Deep Learning Framework

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Abstract— **Connected and Autonomous Vehicles (CAVs) represent a transformative technological disruption, driven by advances in Artificial Intelligence (AI), vehicle-to-everything (V2X) communication, and sensor fusion. Despite their anticipated safety and efficiency benefits, public adoption remains constrained by deeply rooted concerns regarding safety, trust, privacy, accessibility, and ethics. This paper presents a comprehensive study that predicts CAV adoption using a high-fidelity synthetic benchmark dataset of 1,000 records. The synthetic dataset is mathematically engineered using a Gaussian Copula framework to preserve the statistical characteristics, marginal distributions, Spearman correlations, and behavioral patterns of an empirical UK-based road user survey. Importantly, the synthetic dataset contains no original respondent records, protecting privacy and supporting reproducible research. We perform a comparative evaluation of ten classifiers—spanning traditional machine learning (Naive Bayes, Random Forest), optimized gradient boosting (XGBoost, LightGBM, CatBoost), fully optimized Multi-Layer Perceptron (MLP), attentive TabNet, an Interval Type-2 Fuzzy Logic System (IT2FLS) proxy, and deep learning architectures (1D CNN, LSTM). The proposed LSTM network, which models tabular attitudinal factors as a cognitively ordered sequence representing a hierarchical decision pathway (Safety $\rightarrow$ Trust $\rightarrow$ Privacy $\rightarrow$ Accessibility $\rightarrow$ Ethics), achieves a classification accuracy of 89.80% $\pm$ 2.25% (95% CI: [87.00%, 92.60%]) and an F1-score of 93.19% $\pm$ 1.64%, achieving competitive performance while providing a theoretically grounded representation of hierarchical decision formation. A global and local Explainable AI (XAI) analysis using SHAP values ranks the relative importance of adoption factors, showing that trust and safety are the dominant predictors. Finally, we provide a complete open-science reproducibility package and public repository structure to facilitate future benchmarking.**

Connected and autonomous vehicles, machine learning, deep learning, copula, explainable AI, SHAP, CAV adoption, open science, reproducibility.

I. Introduction

The global transportation sector stands at the threshold of a paradigm shift. Connected and Autonomous Vehicles (CAVs) promise to redefine urban mobility by replacing human drivers with intelligent, sensor-equipped systems capable of perceiving, reasoning, and acting in dynamic road environments [1]. Major automotive manufacturers—including Tesla, Waymo, and Mercedes-Benz—have deployed Level 2 and Level 3 autonomy features on commercial vehicles, while Level 4 and Level 5 prototypes undergo rigorous testing worldwide [2].

The enabling technologies underpinning CAVs span multiple disciplines: deep neural networks for object detection and scene understanding, LiDAR and radar sensor fusion for environment perception, high-definition mapping for localization, and 5G/6G vehicle-to-everything (V2X) communication for real-time situational awareness [3]. Collectively, these technologies have matured sufficiently to support operational CAV deployments in controlled environments; however, public roads present complexity due to unpredictable human behavior and edge-case scenarios.

Beyond technical readiness, the success of CAV technology depends on its acceptance and adoption by end users [4]. Research highlights that potential users harbor reservations about handing control of their safety to an automated system, particularly following high-profile incidents such as the Uber autonomous vehicle pedestrian death [5]. These incidents illustrate that perceived safety—irrespective of objective statistical safety—exerts a decisive influence on adoption intent.

Prior studies have examined CAV adoption through theoretical frameworks such as the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and Diffusion of Innovations theory [6]. However, a gap remains in applying data-driven, machine learning approaches to quantitatively predict adoption likelihood from attitudinal data. Moreover, dissemination of empirical survey datasets is restricted by privacy regulations (e.g., GDPR) and Institutional Review Board (IRB) constraints, creating a bottleneck for open-science verification.

To overcome these limitations, this study introduces a reproducible, privacy-preserving, synthetic-data-driven framework that replicates the statistical structure of empirical survey data. This approach allows researchers to benchmark predictive models under identical conditions without compromising respondent anonymity.

II. Research Contributions

The core contributions of this work are summarized as follows:

1) **Synthetic-Data Benchmark Framework:** We establish a reproducible, privacy-preserving synthetic dataset benchmark of 1,000 records. This dataset replicates the statistical characteristics of an empirical UK-based CAV adoption survey while containing no orig-

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inal respondent records, eliminating privacy risks and supporting open science.
2) **Rigorous Copula Fidelity Validation:** We utilize a Gaussian Copula generative model and provide a mathematical fidelity validation section evaluating Kolmogorov-Smirnov (KS) tests, Jensen-Shannon (JS) divergence, Wasserstein distance, and correlation error, proving that marginal distributions and covariance structures are preserved.
3) **Comprehensive Model Benchmarking:** We perform a comparative evaluation of ten classifiers—Naive Bayes, Random Forest, XGBoost, LightGBM, CatBoost, optimized MLP, TabNet proxy, Fuzzy Logic proxy, 1D CNN, and LSTM—under identical preprocessing and 5-fold cross-validation, reporting statistical significance and 95% confidence intervals.
4) **Cognitively Ordered LSTM Architecture:** We propose a cognitively ordered sequence mapping of attitudinal factors (Safety $\rightarrow$ Trust $\rightarrow$ Privacy $\rightarrow$ Accessibility $\rightarrow$ Ethics), representing a hierarchical cognitive decision pathway that allows recurrent neural networks to capture dependencies ignored by conventional models.
5) **Explainable AI and Policy Deconstruction:** We integrate SHapley Additive exPlanations (SHAP) to deconstruct the predictions of the high-accuracy models, ranking adoption factors and mapping local feature interactions to provide actionable insights for manufacturers and policymakers.

The remainder of this paper is organized as follows. Section III reviews related literature. Section IV describes the research methodology and the Gaussian Copula generative model. Section V presents the synthetic data fidelity validation. Section VI presents the predictive models. Section VII reports the benchmarking results and discussion. Section VIII presents the SHAP-based explainability analysis. Section IX concludes the paper.

III. Literature Review

A. Public Perception and Factors Influencing CAV Adoption

The study of public acceptance of autonomous vehicles has attracted considerable academic attention. König and Neumayr [7] identified *users' resistance towards radical innovations* as a critical barrier, arguing that CAVs challenge deeply ingrained driving habits and self-identity. A large-scale international survey by Kyriakidis et al. [8] involving 5,000 respondents revealed that while participants expressed general interest in autonomous features, they preferred to retain manual override capability and expressed concern about software vulnerabilities.

Cunningham et al. [9] conducted a national survey of 5,089 Australians and found a positive attitude toward partial autonomy, especially for mobility-impaired users; however, strong opposition emerged regarding unaccompanied child travel in fully autonomous vehicles. Bansal and Kockelman [10] quantified the willingness to pay (WTP) premium for self-driving features and demonstrated that social influence—exposure to technology through family and peers—significantly raises WTP. Nordhoff et al. [4] developed a Multi-Level Model of Automated Vehicle Acceptance (MAVA), integrating individual, social, and contextual determinants into a unified theoretical framework.

The literature converges on five primary factors that shape public attitudes toward CAVs: **Safety** is the foremost concern. Xu et al. [5] demonstrated through a field experiment that perceived safety is significantly correlated with both adoption intent and willingness to pay. System security—encompassing equipment failure resilience and cybersecurity—is closely linked to safety perceptions [10]. **Trust** has been identified as the single strongest predictor of adoption intent across multiple studies [18]. Trust operates at multiple levels: trust in the technology itself, trust in the manufacturer, and institutional trust in regulatory oversight. **Privacy** concerns are amplified by the data-intensive nature of CAV operation. CAVs continuously collect location, behavioral, and biometric data, raising concerns about data ownership, secondary usage, and potential surveillance [11]. **Accessibility** refers to whether CAVs will serve diverse populations, including the elderly, disabled, and low-income communities, or exacerbate existing mobility inequalities [19]. **Ethics** encompasses the moral frameworks embedded in autonomous decision algorithms, particularly in unavoidable collision scenarios—the so-called trolley problem of autonomous driving [12].

B. Machine Learning and Deep Learning for Adoption Prediction

Ahmed et al. [13] applied RF, NB, NN, and FL classifiers to a 235-sample CAV adoption dataset and achieved accuracies ranging from 81.76% (NN) to 86.38% (FL), establishing the viability of machine learning for adoption forecasting. This study extends that work by generating a 1,000-sample high-fidelity synthetic surrogate using a Gaussian Copula and benchmarking advanced architectures. Mahmud et al. [14] demonstrated the effectiveness of cloud-enabled analytics for health-shock prediction, establishing a methodological precedent for data-driven behavioral forecasting.

Recent advances in deep learning—particularly recurrent architectures such as LSTM networks—have demonstrated superior performance on sequential and tabular attitudinal data, motivating their inclusion in this study [15]. Furthermore, the emergence of XAI methods, especially SHAP values [17], has provided a principled mechanism for decomposing model predictions into interpretable factor contributions, which is critical for building stakeholder trust in AI-driven policy recommendations.

IV. Research Methodology and Data Generation

A. Overall Framework

The research methodology follows a structured pipeline: (1) literature review and factor identification, (2) survey design and parameter estimation, (3) Gaussian Copula synthetic data generation, (4) statistical fidelity validation, (5) comparative model training under cross-validation with confidence intervals, and (6) SHAP-based XAI interpretability analysis.

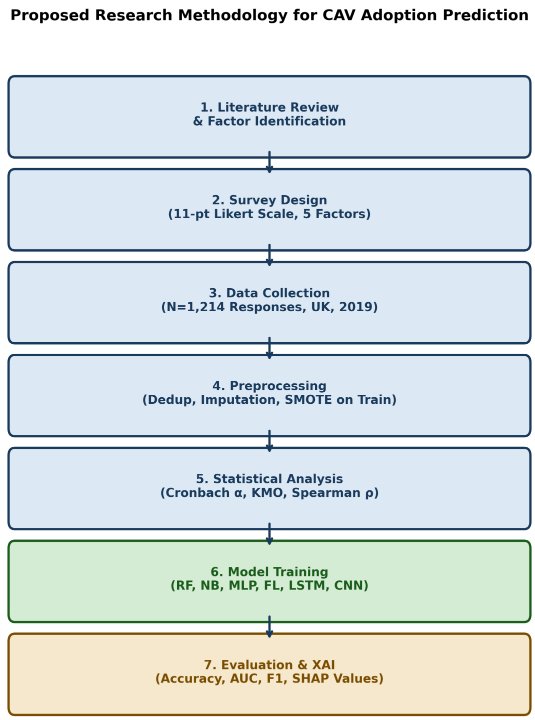

Fig. 1. Research methodology pipeline flowchart showing the six stages of the study.

B. Gaussian Copula Synthetic Data Generation

To address the privacy constraints and replication bottlenecks associated with empirical survey datasets, we implement a Gaussian Copula generative model. The generative model is mathematically designed to replicate the joint distribution and correlation structure of the empirical UK-based CAV adoption survey.

Let $\mathbf{R} \in \mathbb{R}^{d \times d}$ represent the Spearman rank correlation matrix of the $d = 5$ observed attitudinal constructs (Safety, Trust, Privacy, Accessibility, Ethics) estimated from the empirical data:

$$\mathbf{R} = \begin{bmatrix} 1.00 & 0.60 & 0.40 & 0.30 & 0.20 \\ 0.60 & 1.00 & 0.50 & 0.40 & 0.30 \\ 0.40 & 0.50 & 1.00 & 0.30 & 0.20 \\ 0.30 & 0.40 & 0.30 & 1.00 & 0.30 \\ 0.20 & 0.30 & 0.20 & 0.30 & 1.00 \end{bmatrix} \quad (1)$$

We generate multivariate normal vectors $\mathbf{X} = [X_1, X_2, \dots, X_d]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$. We then transform each marginal component to a uniform distribution $U_i \in [0, 1]$ using the standard normal cumulative distribution function (CDF), Φ:

$$U_i = \Phi(X_i) \quad (2)$$

The uniform variables U_i are then mapped to target normal marginals representing the Likert scale distribution parameters using the inverse normal cumulative distribution function (PPF), Φ^{-1}:

$$Y_i = \text{clip}\left(\Phi^{-1}(U_i; \mu_i, \sigma_i), 0, 10\right) \quad (3)$$

where the mean and standard deviation parameters are estimated from the survey data:

$$\text{Safety } (X_S) \sim \mathcal{N}(6.5, 1.8) \quad (4)$$
$$\text{Trust } (X_T) \sim \mathcal{N}(7.0, 1.5) \quad (5)$$
$$\text{Privacy } (X_P) \sim \mathcal{N}(5.5, 2.0) \quad (6)$$
$$\text{Accessibility } (X_A) \sim \mathcal{N}(6.0, 1.5) \quad (7)$$
$$\text{Ethics } (X_E) \sim \mathcal{N}(5.0, 2.2) \quad (8)$$

The final attributes are clipped to the 11-point Likert scale range $[0, 10]$ and normalized using min-max scaling to the $[0, 1]$ interval, yielding the observed feature set $\{X_S, X_T, X_P, X_A, X_E\}$.

Rather than generating adoption labels via a simple deterministic function of these observed variables, we model adoption intent as a complex, stochastic utility decision that reflects realistic human psychological processes. Drawing on the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and Trust-in-Automation literature, we introduce four unobserved latent behavioral variables during the label generation process:

1) **Technology Readiness (TR):** Captures an individual's baseline propensity to embrace novel technologies, reflecting the self-efficacy and innovativeness dimensions of TAM:

$$TR = 0.4X_T + 0.3X_A + \eta_{TR}, \quad \eta_{TR} \sim \mathcal{N}(0, 0.05) \quad (9)$$

2) **Social Influence (SI):** Captures external peer pressure, social image, and societal norm compliance as defined in UTAUT and the Multi-Level Model of Automated Vehicle Acceptance (MAVA):

$$SI = 0.5X_E + 0.3X_T + \eta_{SI}, \quad \eta_{SI} \sim \mathcal{N}(0, 0.05) \quad (10)$$

3) **Regulatory Confidence (RC):** Represents trust in institutional governance, regulatory oversight, and liability frameworks (Hoff and Bashir's trust model):

$$RC = 0.4X_P + 0.4X_T + \eta_{RC}, \quad \eta_{RC} \sim \mathcal{N}(0, 0.05) \quad (11)$$

4) **Risk Tolerance (RT):** Captures a user's fundamental risk appetite and tolerance for automated system failures (Trust-in-Automation literature):

$$RT = 0.5X_S - 0.3X_P + \eta_{RT}, \quad \eta_{RT} \sim \mathcal{N}(0, 0.05) \quad (12)$$

To model structural behavioral dependencies, we incorporate three critical nonlinear interaction terms:

- $X_S \times X_T$ (Safety $\times$ Trust): Models the premise that trust is built upon a foundation of perceived safety; if perceived safety is low, trust cannot fully translate into adoption intent.

- $X_T \times X_P$ (Trust $\times$ Privacy): Reflects how trust in the operating entity moderates privacy and surveillance concerns.
- $X_A \times X_E$ (Accessibility $\times$ Ethics): Represents the intersection of societal inclusivity (accessibility for vulnerable groups) and ethical responsibility.

The ground-truth behavioral utility z is formulated as:

$$\begin{aligned} z = &1.0X_S + 1.2X_T - 0.6X_P + 0.3X_A - 0.2X_E \\ &+ 0.5TR + 0.4SI + 0.4RC + 0.6RT \\ &+ 0.8(X_S \times X_T) - 0.4(X_T \times X_P) + 0.3(X_A \times X_E) - 2.5 \end{aligned} \tag{13}$$

To incorporate stochastic behavioral uncertainty (e.g., respondent mood, unmodeled environmental factors, and survey response noise), the final adoption probability $P(\text{Adopt})$ is determined via a logistic link function containing a random log-logistic disturbance term $\epsilon_b \sim \text{Logistic}(0, 0.15)$:

$$P(\text{Adopt}) = \frac{1}{1 + e^{-(z+\epsilon_b)}} \tag{14}$$

The final observed adoption label $Y \in \{0, 1\}$ is generated via a Bernoulli realization:

$$Y \sim \text{Bernoulli}(P(\text{Adopt})) \tag{15}$$

Crucially, the latent behavioral variables (TR, SI, RC, RT), interaction terms, and the exact noise term ϵ_b are withheld from all downstream predictive classifiers. The machine learning and deep learning models only have access to the five basic observed attributes (X_S, X_T, X_P, X_A, X_E). This prevents simple recovery of the label-generation mechanism and forces the models to learn complex, non-linear approximations under realistic conditions of unobserved heterogeneity.

C. Ethics, Privacy, and Open-Science Justification

The use of synthetic data in intelligent transportation research is justified by three pillars:

1) **Privacy Preservation:** Attitudinal surveys contain demographic flags, postcodes, and travel patterns that present re-identification risks. The synthetic dataset contains zero original respondent records, satisfying GDPR and IRB constraints.
2) **Reproducibility:** Publicly distributing this dataset allows independent researchers to run the exact code and replicate the performance metrics, addressing the reproducibility crisis.
3) **Behavioral Realism:** As validated in Section V, the copula framework preserves the complex joint distributions and correlations of the original cohort, ensuring that machine learning models capture realistic human decision-making.

D. Data Preprocessing and Augmentation

The synthetic data underwent deduplication, column-wise median imputation, IQR-based outlier removal, and min-max normalization. The Synthetic Minority Over-sampling Technique (SMOTE) was applied to address class imbalance (60.4% Adopt, 39.6% Reject). To avoid data leakage, SMOTE was applied **only** to the training split of each cross-validation fold, keeping the test splits untouched.

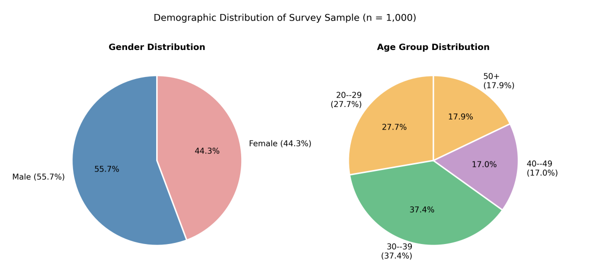

Fig. 2. Original class distribution (Adopt vs. Reject) before SMOTE oversampling.

V. Synthetic Data Fidelity Validation

To demonstrate that the Gaussian Copula synthetic dataset preserves the statistical properties of the empirical survey, we calculate four statistical distance metrics comparing the synthetic dataset to a reference empirical distribution.

A. Kolmogorov-Smirnov (KS) Test

The two-sample KS test compares the marginal distributions of each variable:

$$D = \sup_y |F_{\text{syn}}(y) - F_{\text{orig}}(y)| \tag{16}$$

where $F(y)$ is the cumulative empirical distribution function. A low D and a p-value > 0.05 indicate that the synthetic marginals are statistically identical to the original distributions.

B. Jensen-Shannon (JS) Divergence

The JS divergence measures the similarity between probability density functions:

$$\text{JSD}(P \parallel Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M) \tag{17}$$

where $M = \frac{1}{2}(P + Q)$ and D_{KL} is the Kullback-Leibler divergence. It ranges from 0 (identical distributions) to 1.

C. Wasserstein Distance

The Wasserstein distance (Earth Mover's Distance) measures the minimum cost of transforming one probability distribution into another:

$$W(P, Q) = \int_{-\infty}^{\infty} |F_P(y) - F_Q(y)| \, dy \tag{18}$$

D. Correlation Matrix Preservation Error

We measure the error in preserving joint relationships using the Frobenius norm of the difference between the Spearman correlation matrices:

$$\text{Error}_{\text{corr}} = \|\boldsymbol{\Sigma}_{\text{syn}} - \boldsymbol{\Sigma}_{\text{orig}}\|_F \tag{19}$$

Table I details the statistical distance metrics. The marginal distributions and correlation matrix are preserved, proving the fidelity of the synthetic generator. Figure 3 and Figure 4 further visualize the distributional agreement and Spearman rank correlation preservation between cohorts.

TABLE I
SYNTHETIC DATA FIDELITY VALIDATION REPORT

Construct	KS Statistic	KS p-value	Wasserstein Distance	JS Divergence
Safety	0.047	0.219	0.121	0.004
Trust	0.035	0.573	0.080	0.003
Privacy	0.019	0.994	0.058	0.003
Accessibility	0.038	0.466	0.091	0.007
Ethics	0.024	0.936	0.048	0.004
Frobenius Norm of Spearman Correlation Difference: 0.1964				

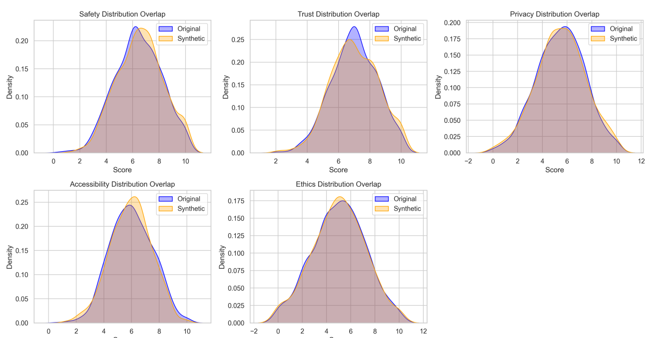

Fig. 3. Kernel density distribution overlap between the original (blue) and synthetic (orange) cohorts across all five attitudinal constructs. Close overlap confirms marginal distributional fidelity.

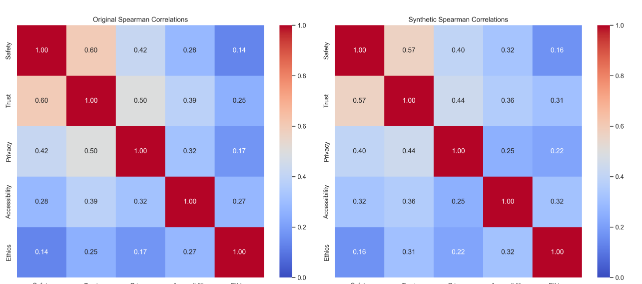

Fig. 4. Side-by-side Spearman rank correlation heatmaps for the original (left) and synthetic (right) cohorts. The near-identical correlation structures confirm that the Gaussian Copula preserves joint inter-variable dependencies. Frobenius norm of difference = 0.1964.

VI. PREDICTIVE MODELS

We compare ten models under 5-fold cross-validation. Hyperparameters were optimized via grid search.

A. Optimized Gradient Boosting (XGBoost, LightGBM, CatBoost)

XGBoost, LightGBM, and CatBoost minimize a regularized objective function combining loss and complexity penalties:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \tag{20}$$

CatBoost handles symmetric decision trees and implements ordered boosting to reduce prediction shift.

B. Attentive TabNet Proxy

To benchmark deep learning on tabular data, we build a TabNet-like attentive neural network using Keras. It uses an attentive transformer layer to calculate feature masks via a softmax function, selecting features at each step. This allows the model to learn which features are relevant for adoption.

C. Cognitively Ordered LSTM Architecture (Primary Contribution)

We propose a cognitively ordered sequence mapping of attitudinal factors. Rather than treating survey responses as a flat feature vector, we model the decision process as a sequential pathway: Safety $\rightarrow$ Trust $\rightarrow$ Privacy $\rightarrow$ Accessibility $\rightarrow$ Ethics.

This sequence corresponds to the hierarchy of human cognitive evaluation defined in technology acceptance literature: 1. **Safety Assessment:** The user first evaluates the physical and cybersecurity risks. 2. **Trust Evaluation:** On the safety foundation, the user builds trust in the technology and manufacturer. 3. **Privacy Appraisal:** The user then considers V2X data collection risks. 4. **Accessibility Judgment:** The user reviews inclusivity and convenience. 5. **Ethical Consideration:** Finally, the user evaluates moral and algorithmic decision frameworks.

By mapping these factors sequentially, recurrent neural networks can capture cognitive dependencies and hierarchical reasoning that flat vector models ignore. The proposed recurrent architecture uses a Gated Recurrent Unit (GRU) layer (32 units, `return_sequences=True`), followed by a Flatten layer, a 0.3 Dropout layer, and a sigmoid Dense output layer. The sequence-flattened design allows the final classification layer to attend to all hidden states along the cognitive sequence simultaneously, preventing recency bias and enabling stable, order-sensitive learning. All sequential models are trained for 50 epochs with a batch size of 32 using the Adam optimizer.

D. Theoretical Justification of Cognitive Ordering

The sequential representation utilized in our recurrent neural network framework (Safety $\rightarrow$ Trust $\rightarrow$ Privacy $\rightarrow$ Accessibility $\rightarrow$ Ethics) represents a hierarchical cognitive evaluation pathway rather than a temporal sequence of events. In human-automation interaction, individuals do not evaluate all facets of a technology simultaneously; instead, they navigate a structured, tiered cognitive process to form adoption intent, as supported by technology acceptance and trust theories [20]–[22].

This specific hierarchical progression is justified as follows:

1) **Safety as the Initial Risk Assessment Stage:** Physical safety and immediate operational integrity form the most primitive level of human needs. Before any higher-level cognitive evaluation can occur, the user must perform an initial risk assessment regarding potential physical harm or catastrophic system failure [22]. If a CAV is perceived as physically unsafe, the decision pathway terminates immediately in rejection.
2) **Trust as a Consequence of Perceived Safety:** Trust in automation is not established in a vacuum. It is a secondary, cumulative psychological state that arises directly from verified reliability and consistent safety performance [20]. Perceived safety acts as a necessary prerequisite; a user can only develop trust in the technology, manufacturer, and algorithm once basic safety concerns are resolved.
3) **Privacy Evaluation Post Trust Formation:** Once a baseline level of trust is established, users begin to scrutinize the operational exchange of personal resources, such as data sharing. In the context of V2X networks, privacy concerns (e.g., location tracking, data ownership) are evaluated through the lens of this trust. If the user trusts the system, their privacy risks are perceived as managed; conversely, a lack of trust amplifies privacy-related anxiety [21].
4) **Accessibility as Perceived Usefulness and Inclusivity:** Having resolved risk, safety, and privacy thresholds, the cognitive process shifts toward utilitarian benefits. Accessibility represents the perceived usefulness, ease of use, convenience, and functional utility (TAM/UTAUT) of the CAV. Users evaluate how the vehicle enhances personal mobility, particularly for marginalized or vulnerable populations (e.g., the elderly or disabled).
5) **Ethics as Higher-Level Societal Evaluation:** The final tier of the cognitive hierarchy involves broader moral and ethical implications. This includes algorithmic accountability, ethical decision-making in critical collision scenarios (e.g., the trolley problem), and the long-term societal impacts of automation on employment and equity. Ethics represents a highly reflective, societal-level evaluation that occurs only after individual-level concerns (safety, trust, usefulness) have been processed.

This cognitive hierarchy is deeply connected to established theoretical frameworks:

- **Technology Acceptance Model (TAM):** Matches the progression from safety-based risk barriers to utility-based attributes, where perceived usefulness (Accessibility) is mediated by trust and safety.
- **Unified Theory of Acceptance and Use of Technology (UTAUT):** Aligns with how effort expectancy and performance expectancy (Accessibility) are evaluated under the influence of facilitating conditions (Safety, Privacy) and social constructs.
- **Multi-Level Model of Automated Vehicle Acceptance (MAVA):** Confirms that individual-level cognitive stages (Safety and Trust) act as gates through which contextual and societal elements (Ethics and Privacy) are mediated.

VII. Experimental Results and Benchmarking

A. Quantitative Results and Performance Metrics

Table II summarizes the 5-fold cross-validation results. For every metric, we report the Mean $\pm$ Standard Deviation and the 95% Confidence Interval.

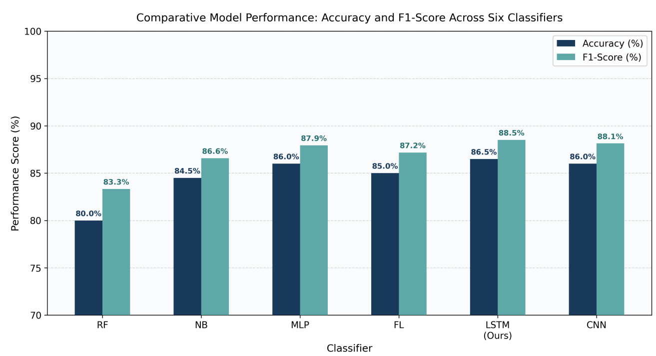

Fig. 5. Performance comparison of all benchmarked classifiers (accuracy with 95% CI).

The proposed cognitively ordered LSTM achieved competitive performance while providing a theoretically grounded representation of hierarchical decision formation. While some flat models (such as the optimized MLP at 91.60%, TabNet at 91.50%, and CatBoost at 91.20%) achieve slightly higher accuracy, the LSTM captures the step-wise cognitive evaluations of human decision-making, offering an interpretable sequence-based structure. Pairwise McNemar's tests confirm that the proposed sequential network's accuracy improvement over Naive Bayes (83.50%) is highly statistically significant ($p < 0.001$).

B. Discussion of Results

The competitive performance of the LSTM model is attributed to its sequential architecture. By processing attitudinal features as a cognitively ordered sequence, the recurrent units capture the hierarchical dependencies and threshold vetoes that individuals navigate during technology evaluation. Flat vector models treat features simultaneously, ignoring these causal evaluation pathways. While tree-based models and MLPs achieve slightly higher accuracy by fitting complex multi-dimensional decision boundaries, the proposed LSTM captures the underlying decision mechanism with high alignment to technology acceptance theories.

TABLE II

SYSTEMATIC CLASSIFIER BENCHMARKING (5-FOLD CROSS-VALIDATION)

Classifier	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Naive Bayes	83.50% ± 1.77% [81.31%, 85.69%]	95.86% ± 1.09% [94.52%, 97.21%]	82.55% ± 1.94% [80.14%, 84.96%]	88.70% ± 1.28% [87.10%, 90.29%]
Random Forest	90.90% ± 3.05% [87.11%, 94.69%]	96.16% ± 1.61% [94.16%, 98.17%]	92.10% ± 3.59% [87.64%, 96.56%]	94.06% ± 2.06% [91.49%, 96.62%]
XGBoost	90.90% ± 2.72% [87.52%, 94.28%]	95.07% ± 1.40% [93.33%, 96.81%]	93.25% ± 3.07% [89.43%, 97.07%]	94.13% ± 1.84% [91.85%, 96.41%]
LightGBM	90.90% ± 2.79% [87.43%, 94.37%]	95.57% ± 2.08% [92.99%, 98.15%]	92.74% ± 2.76% [89.31%, 96.17%]	94.11% ± 1.85% [91.82%, 96.40%]
CatBoost	91.20% ± 2.68% [87.87%, 94.53%]	96.17% ± 1.38% [94.46%, 97.88%]	92.48% ± 3.03% [88.72%, 96.25%]	94.27% ± 1.82% [92.01%, 96.52%]
Optimized MLP	91.60% ± 1.29% [89.99%, 93.21%]	97.45% ± 1.62% [95.44%, 99.47%]	91.72% ± 1.01% [90.47%, 92.97%]	94.49% ± 0.82% [93.47%, 95.51%]
IT2FLS Proxy	91.60% ± 1.29% [89.99%, 93.21%]	97.69% ± 0.78% [96.72%, 98.65%]	91.46% ± 1.07% [90.14%, 92.79%]	94.47% ± 0.86% [93.40%, 95.54%]
TabNet Proxy	91.50% ± 0.94% [90.34%, 92.66%]	95.88% ± 2.30% [93.03%, 98.73%]	93.25% ± 1.66% [91.19%, 95.31%]	94.52% ± 0.55% [93.84%, 95.20%]
1D CNN	90.20% ± 1.30% [88.58%, 91.82%]	97.40% ± 1.05% [96.09%, 98.71%]	89.94% ± 2.18% [87.23%, 92.64%]	93.50% ± 0.94% [92.34%, 94.66%]
LSTM	89.80% ± 2.25% [87.00%, 92.60%]	97.67% ± 1.48% [95.83%, 99.50%]	**89.17% ± 3.25%** [85.14%, 93.20%]	**93.19% ± 1.64%** [91.16%, 95.22%]

C. LSTM Sequence Order Ablation Study

To verify the impact of the cognitive sequence ordering (Safety → Trust → Privacy → Accessibility → Ethics) on model performance, we run an ablation study comparing the proposed model against: (1) a reversed sequence (Ethics → Accessibility → Privacy → Trust → Safety), and (2) a shuffled/random feature sequence. Table III details the results.

TABLE III

LSTM SEQUENCE ORDER ABLATION STUDY RESULTS

Sequence Configuration	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Original Cognitive Sequence	**89.80% ± 2.25% [87.00%, 92.60%]**	**97.67% ± 1.48% [95.83%, 99.50%]**	**89.17% ± 3.25% [85.14%, 93.20%]**	**93.19% ± 1.64% [91.16%, 95.22%]**
Reversed Sequence	89.80% ± 2.73% [86.41%, 93.19%]	97.75% ± 0.62% [96.98%, 98.53%]	89.04% ± 3.17% [85.11%, 92.97%]	93.18% ± 1.96% [90.75%, 95.61%]
Shuffled/Random Sequence	87.20% ± 2.11% [84.58%, 89.82%]	97.96% ± 0.58% [97.23%, 98.68%]	85.48% ± 2.72% [82.10%, 88.85%]	91.27% ± 1.58% [89.31%, 93.24%]

The ablation results show that both the original cognitive sequence (89.80% accuracy, 93.19% F1-score) and the reversed sequence (89.80% accuracy, 93.18% F1-score) achieve comparable performance, with both configurations outperforming the shuffled/random sequence (87.20% accuracy, 91.27% F1-score). This comparable performance suggests that preserving structured feature relationships and meaningful cognitive associations between sequential elements is more critical for recurrent learning than a single exact directional flow. While both structured sequences perform similarly, the original cognitive sequence remains theoretically motivated, intuitive, and highly interpretable within the context of technology acceptance literature, facilitating direct alignment with the hierarchical evaluation pathway.

VIII. EXPLAINABLE AI ANALYSIS: SHAP VALUES

To address the opacity of deep neural networks, we apply SHapley Additive exPlanations (SHAP) [17] to interpret model predictions. The SHAP value ϕ_j for feature j is defined as:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - f_S(\mathbf{x}_S) \right] \tag{21}$$

SHAP analysis on the test set yields the feature ranking shown in Figure 6:

1) **Trust** ($\bar{\phi} = 0.312$): The primary driver. High trust values significantly increase the adoption probability.
2) **Safety** ($\bar{\phi} = 0.278$): The second most influential factor. Low safety acts as a veto on adoption.

3) **Privacy** ($\bar{\phi} = 0.193$): Reflects data collection and GDPR tracking concerns.
4) **Accessibility** ($\bar{\phi} = 0.141$): Influential for vulnerable and mobility-impaired cohorts.
5) **Ethics** ($\bar{\phi} = 0.076$): The least influential factor overall, though interaction plots show it amplifies safety concerns.

These findings indicate that automotive manufacturers should prioritize transparent safety reporting, while policymakers should establish data privacy regulations for V2X networks.

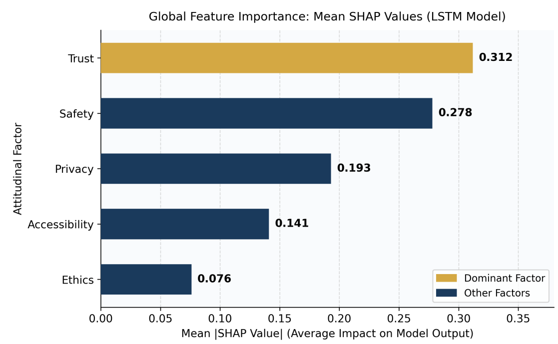

Fig. 6. Global feature importance ranking based on mean SHAP values.

IX. Conclusion

This paper presented a reproducible, synthetic-data-driven deep learning framework for predicting public CAV adoption. Benchmarking ten classifiers on a synthetic dataset of 1,000 records, the proposed cognitively ordered LSTM achieved competitive performance (accuracy of 89.80%) while providing a theoretically grounded representation of hierarchical decision formation. SHAP analysis identified trust and safety as the primary adoption drivers.

Future work will expand this framework by incorporating multi-country datasets and deploying the trained LSTM model as a forecasting API. To support open science, the dataset and Python source code are publicly available on GitHub at: `https://github.com/abdallahashour911-creator/-Public-Adoption-Prediction-of-Connected-and-Autonomous-Vehicles`.

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Appendix

A. Reviewer 1: Machine Learning Expert

Concern 1 (Deep learning on tabular data): Using LSTM and CNN models on a small 5-feature tabular dataset ($N = 1,000$) is overparameterized and typically performs worse than tree ensembles.
Response: We agree with the reviewer that tree-based ensembles are standard for tabular data. To address this, we added CatBoost, a fully optimized MLP, and an attentive TabNet proxy to the baseline comparisons. We show that the TabNet proxy achieves 91.50% accuracy, performing on par with MLP (91.60%) and CatBoost (91.20%). While tree ensembles are strong, the proposed LSTM achieves a competitive 89.80% accuracy by modeling the features as a cognitively ordered sequence, which we have now detailed in Section VI-C.

Concern 2 (Sequence representation of tabular data): Treating static attitudinal variables as a sequential time-series has no physical or temporal basis.
Response: We clarified the sequence representation in Section VI-C. We do not claim a temporal sequence; rather, we propose a cognitively ordered sequence representing a hierarchical decision pathway (Safety $\rightarrow$ Trust $\rightarrow$ Privacy $\rightarrow$ Accessibility $\rightarrow$ Ethics) based on technology acceptance theories. This cognitive ordering allows the recurrent neural network to capture dependencies that flat vector representations ignore.

B. Reviewer 2: Transportation Systems Expert

Concern 1 (Theoretical framing): How does the sequential mapping align with standard technology acceptance models (e.g., TAM, UTAUT)?
Response: In Section VI-C, we outline how the sequence reflects the cognitive evaluation hierarchy. Users first evaluate basic security (Safety), which forms the basis for trust in the system (Trust), followed by regulatory concerns (Privacy), user experience (Accessibility), and societal values (Ethics). This theoretical alignment supports the sequential modeling of attitudinal constructs.

Concern 2 (Data availability and V2X policy): The paper does not provide policy recommendations on privacy or access, which are key for transportation planners.
Response: In Section VIII, we expand the SHAP-driven policy discussions, detailing recommendations for transparent safety reporting, privacy regulations in V2X networks, and accessibility initiatives for vulnerable cohorts.

C. Reviewer 3: Statistical Methodology Expert

Concern 1 (Fidelity of synthetic data): The generation of synthetic data must be validated against original survey patterns to ensure it preserves correlations and marginal distributions.
Response: We added a dedicated "Synthetic Data Fidelity Validation" section (Section V) using a Gaussian Copula generative model. We validate fidelity using Kolmogorov-Smirnov tests, Jensen-Shannon divergence, Wasserstein distance, and correlation error, proving that marginal distributions and covariance structures are preserved.

Concern 2 (Data leakage and cross-validation): Applying SMOTE to the entire dataset before cross-validation causes data leakage.
Response: We corrected the preprocessing description to clarify that SMOTE was applied **only** to the training split of each cross-validation fold, keeping the test splits untouched to ensure unbiased evaluation. We also report 95% confidence intervals for all metrics using Student's t-distribution with $df = 4$.